PES UNIVERSITY • DEPT. OF COMPUTER SCIENCE AND ENGINEERING

Lab 2: Reflection Questions

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Problem Statement: #43 – Feature Flag & Dynamic Config Manager

Course: Software Engineering Laboratory (Lab 2)

Reflection Questions & Technical Analysis

1. Did your estimations reflect the actual effort?

Yes. Relative estimation using the Fibonacci sequence accurately captured development complexity, technical risk, and testing effort. Low-uncertainty CRUD items (e.g. US-01, 3 pts) progressed quickly without blockers. Algorithmic stories (e.g. US-02 deterministic hashing and US-09 cryptographic audit logging, 5 pts each) accounted for statistical cohort validation and transaction integrity testing. High-risk performance engineering (US-04 sub-15ms evaluation latency, 8 pts) correctly mirrored the iterative profiling, memory caching, and benchmark tuning needed to fulfill NFR-001. Abstracting effort into story points rather than calendar hours protected the team from false precision and Parkinson's Law.

2. Was your backlog well-prioritized?

Yes. Backlog prioritization followed architectural dependency sequencing and risk-reduction principles. Core flag creation (US-01) and deterministic evaluation (US-02, US-03) were scheduled at the top of Sprint 1 because downstream approval gates and WebSocket sync require a stable evaluation core to operate upon. Furthermore, scheduling the high-risk performance optimization (US-04) in Sprint 1 eliminated architectural risk early. Enterprise governance, environment isolation, and real-time streaming were layered cleanly in Sprint 2, creating a resilient MVP progression.

3. How did your simulated sprint align with your plan?

The simulated sprints closely matched planned execution. In Sprint 1, 24 planned points were delivered across 5 user stories with zero scope creep. In Sprint 2, all 23 points were completed. Pacing was disciplined: smaller 3-point stories moved to Done early in both sprints, while complex 5-point and 8-point items progressed through comprehensive unit and load verification before transitioning to Done by Day 5. All sprint boundaries were protected against unplanned scope injections.

4. What insights did the burndown chart give about your team's capacity?

Jira's Burndown Chart revealed three key operational insights: (1) Predictable Capacity: Completing 24 points in Sprint 1 and 23 points in Sprint 2 established that the team has a reliable, sustainable velocity of ~23–24 points per 1-week sprint. (2) Story Sizing & Burndown Trajectory: Large work items (e.g. 8-point US-04) create step-like burndown plateaus because points only burn upon complete verification. Decomposing 8-point stories into smaller 3-point slices in future sprints will yield a smoother burndown curve. (3) Discipline & Zero Scope Creep: The absence of upward spikes on the remaining values line confirmed strict sprint backlog discipline without uncommitted task creep.